Battery Management System

Abstract

This project presents an innovative system for automatically charging the battery of Electric Vehicles (EVs) by harnessing energy from the rotation of the car wheels. The concept builds upon regenerative braking principles but extends it to continuous energy harvesting during driving. Integrated with a sophisticated Battery Management System (BMS), the system monitors key parameters including voltage, SOC, and SOH in real-time. The goal is to extend driving range, improve energy efficiency, and enhance battery longevity. Challenges related to energy conservation and mechanical losses are addressed through optimized control strategies.

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Chapter 3: System Design

Block diagram, generators, power electronics, sensors for voltage/current/temperature. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery.

BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics).

Block diagram, generators, power electronics, sensors for voltage/current/temperature. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery.

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Block diagram, generators, power electronics, sensors for voltage/current/temperature. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery.

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Block diagram, generators, power electronics, sensors for voltage/current/temperature. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery.BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics).

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Block diagram, generators, power electronics, sensors for

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Block diagram, generators, power electronics, sensors for voltage/current/temperature. Wheel rotation energy harvesting supplements battery.

BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery.
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BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics).

Chapter 4: Battery Management System

Detailed BMS functions, protection, balancing, and real-time monitoring. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery.
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Detailed BMS functions, protection, balancing, and real-time monitoring. Wheel

rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery.
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Chapter 5: Implementation

Hardware, software, dashboard for voltage, SOC, SOH. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage
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Hardware, software, dashboard for voltage, SOC, SOH. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage
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Chapter 6: Testing and Results

Simulations, performance metrics, graphs. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage
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Chapter 7: Conclusion

Summary, benefits, future scope. Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics). Wheel rotation energy harvesting supplements battery. BMS tracks voltage (3.0-4.2V/cell), SOC (0-100%), SOH (capacity and resistance metrics).

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Parameter	Description	Value
Voltage	Cell level monitoring	3.0 - 4.2 V
SOC	State of Charge	Estimated via algorithms
SOH	State of Health	Degradation tracking